Grade 10 Term 1 Lesson 1 Practice Activity 2
Revenue, Costs, Profit, and Payoff Tables

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Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C3 Calculates profit as revenue minus costs across states of nature and organises the resulting payoffs in payoff tables.

1 Introduction

Problems:

1.1. From a decision situation to a profit table

1.2. Introductory General Problem (C1–C3)

1.3. Build totals from quantities, rates, and fixed amounts

1.4. Advanced General Problem (C1–C3)

1.5. Reference and classroom solving routine

1.1 From a decision situation to a profit table

Start with the situation

A school enterprise organises services for community events. Before bookings are confirmed, its students must choose either a basic service plan or an expanded service plan with more equipment and capacity. Demand may turn out to be high or low. The enterprise chooses its plan first; demand is revealed later. This timing makes the decision uncertain.

The uncertain process is customers making bookings. High demand and low demand are the two possible **states** of that process in this model. The students control the service plan, but they cannot choose the demand state. A consequence can first be described in words: "the basic plan operates under high demand, earns sales income, and uses staff and supplies." We then assign a monetary payoff to that consequence. Here the payoff will be the profit remaining after costs. Describing what happens and measuring its financial result are related, but different, tasks.

Name the choices, then the states

The basic and expanded plans are the **alternatives**: the available choices. We now shorten their names to A_1, read "alternative one," and A_2, read "alternative two."

Symbol	Choice made by the enterprise
A_1	Choose the basic service plan.
A_2	Choose the expanded service plan.

High and low demand are possible outcomes outside the enterprise's control. We label them S_1, read "state one," and S_2, read "state two."

Symbol	Demand outcome revealed after the choice
S_1	High demand.
S_2	Low demand.

The small numbers below the letters are **subscripts**. They identify choices and states; they do not rank them or give their monetary values.

Check. Can the enterprise choose S_1? *Suggested answer:* No. It chooses a plan; customers' bookings determine which demand state occurs.

Match one choice with one state

To describe a possible result, name both the chosen plan and the demand outcome. The pair (A_1, S_1), read "alternative one with state one," means the basic plan under high demand. There are four combinations:

Pair	Contextual meaning
(A_1, S_1)	Basic plan, high demand.
(A_1, S_2)	Basic plan, low demand.
(A_2, S_1)	Expanded plan, high demand.
(A_2, S_2)	Expanded plan, low demand.

Only one pair will be realised for this particular decision. We analyse all four beforehand so that the financial consequence of each possible choice–state combination is recorded.

Give the consequence a monetary value

Revenue is money earned from the activity. **Cost** is money spent or resources charged to the activity. **Profit** is revenue minus cost. Suppose the basic plan under high demand earns 20 thousand USD and costs 9 thousand USD. Its profit is $20 - 9 = 11$ thousand USD. One thousand USD means 1,000 US dollars. Throughout these introductory examples, monetary totals are in thousand USD. Revenue alone is not the payoff in this activity: profit is.

We introduce a separate symbol for each total. First,

$$R(A_1, S_1) = 20.$$

Read this as "total revenue for alternative one in state one is 20 thousand USD." The capital R names total revenue; A_1 identifies the basic plan and S_1 identifies high demand. The parentheses identify the inputs to this quantity; they do not mean multiplication.

Next, write the corresponding total cost as

$$C(A_1, S_1) = 9.$$

Read this as "total cost for alternative one in state one is 9 thousand USD." Capital C names total cost, with the same two inputs.

Finally, capital P names profit, which is our payoff:

$$P(A_1, S_1) = R(A_1, S_1) - C(A_1, S_1) = 11.$$

Read this as "profit for alternative one in state one equals its revenue minus its cost: 11 thousand USD." All three quantities refer to the *same* plan and demand state. The result is positive because revenue exceeds cost. In separate small illustrations, revenue 9 and cost 9 give zero profit $(9 - 9 = 0)$; revenue 8 and cost 9 give a loss $(8 - 9 = -1$ thousand USD).

Organise the four profits

A profit payoff table puts alternatives in rows and states in columns. For example, take the basic plan's high-demand revenue of 20 and matching cost of 9, subtract to obtain 11, then put 11 in row A_1, column S_1. Repeat that process for the other three pairs. The next problem consolidates this process using directly supplied totals. Its input table records the data; the paired calculation tables match each symbolic subtraction to its arithmetic; the clean final table contains profits only.

Check. Could we subtract the cost for (A_2, S_1) from the revenue for (A_1, S_1)? *Suggested answer:* No. That would mix two plans. Both amounts in a subtraction must belong to the same alternative–state pair.

1.2 Introductory General Problem (C1–C3)

Use the school service enterprise just described. It must commit to either alternative A_1 or alternative A_2 before learning whether demand will be high (S_1) or low (S_2). Its accounting team used sales forecasts for each choice under each demand outcome to estimate total revenue, and used purchasing and operating budgets to estimate the corresponding total cost. All amounts and the resulting profit are measured in thousand USD.

If the enterprise selects A_1, forecast sales produce total revenue of 20 under demand state S_1 and 13 under S_2; fulfilling those sales is budgeted to cost 9 and 8, respectively. If it selects A_2, its different sales plan is expected to bring in 18 under S_1 and 15 under S_2, while the resources required by that plan cost 10 and 9, respectively. The enterprise must calculate the profit for every possible alternative–state pair before constructing its payoff table.

Questions/Tasks.

C1 Interpret the decision elements.

C3 Build the payoff table.

C1 - Interpreting the decision.

Element	Description	
Alternatives	A_1**:** Basic service plan	A_2**:** Expanded service plan
States of nature	S_1**:** High demand	S_2**:** Low demand
Uncertain event	Customer bookings reveal demand after the enterprise chooses its plan	
Consequences	Services are delivered with sales income and resource use; profit measures the monetary consequence in thousand USD	



C3 - Calculating revenue, cost, and payoff.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Operating alternative 1.
A_2	—	Operating alternative 2.
S_1	—	Demand outcome 1.
S_2	—	Demand outcome 2.
$R(A_1, S_1)$	20 thousand USD	Forecast total revenue for A_1 under S_1.
$C(A_1, S_1)$	9 thousand USD	Budgeted total cost for A_1 under S_1.
$R(A_1, S_2)$	13 thousand USD	Forecast total revenue for A_1 under S_2.
$C(A_1, S_2)$	8 thousand USD	Budgeted total cost for A_1 under S_2.
$R(A_2, S_1)$	18 thousand USD	Forecast total revenue for A_2 under S_1.
$C(A_2, S_1)$	10 thousand USD	Budgeted total cost for A_2 under S_1.
$R(A_2, S_2)$	15 thousand USD	Forecast total revenue for A_2 under S_2.
$C(A_2, S_2)$	9 thousand USD	Budgeted total cost for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$20 - 9$ $= 11$	$13 - 8$ $= 5$
A_2	$18 - 10$ $= 8$	$15 - 9$ $= 6$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	11	5
A_2	8	6



1.3 Build totals from quantities, rates, and fixed amounts

Generalise a familiar calculation

The first problem has two choices and two states. To refer to any one of them without writing a separate rule for every cell, use A_i, read "alternative i," and S_j, read "state j." The letters i and j are **indices**: they tell us which alternative and which state to use. For example, substituting $i = 2$ and $j = 1$ gives A_2, S_1, and the expanded-plan, high-demand pair (A_2, S_1). In the first problem, $R(A_2, S_1) = 18$, $C(A_2, S_1) = 10$, and $P(A_2, S_1) = 18 - 10 = 8$ thousand USD. The indices 2 and 1 are identifiers; the number 8 is a monetary value. The number of alternatives is a count, not an index or a profit. Later problems may have more choices or states.

The relationship already used in every cell can now be written generally:

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

This says "profit for a chosen alternative and a particular state is the revenue for that pair minus the cost for that same pair."

Why build revenue and cost?

A **direct model**, such as the first problem, supplies total revenue and total cost. Use those totals as given; we cannot infer a unique set of prices, quantities, or fixed payments from them. A **component model** instead supplies ingredients: quantities, rates per unit, and fixed amounts. We must use those ingredients to build total revenue and total cost *independently*, then subtract.

We now study a new version of the same school service enterprise. Its plans and demand labels keep their meanings, but its financial arrangements will be stated component by component. These are new assumptions, not a breakdown of the first problem's directly supplied totals. Each extension below explicitly adds an arrangement to this new version.

Quantity: how much service is completed?

Suppose the basic plan completes two service batches under high demand and one under low demand. A **service batch** is one package of work for an event; it is the quantity unit throughout this running example. The expanded plan can complete three batches under high demand and two under low demand. Write $q(A_1, S_1) = 2$ batches, read "quantity for alternative one in state one is two batches." More generally, $q(A_i, S_j)$ means the quantity for the named pair.

Quantity (service batches)	S_1: high demand	S_2: low demand
A_1: basic plan	2	1
A_2: expanded plan	3	2

Demand affects bookings, while the chosen plan affects capacity, so quantity can depend on both inputs. A quantity of 2 batches is not 2 thousand USD.

A revenue rate determined by the plan

Assume initially that the only revenue is payment for each completed batch. The basic plan earns 10 thousand USD per batch; the expanded plan earns 11 thousand USD per batch for its larger service package. These are earning **rates**, rather than total amounts. Write $r(A_1) = 10$ and $r(A_2) = 11$ thousand USD per batch. Read $r(A_1)$ as "revenue rate for alternative one." More generally, $r(A_i)$ is the rate set by the plan.

Multiply the number of batches by the payment for each batch:

$$R(A_i, S_j) = q(A_i, S_j) r(A_i).$$

For the basic plan under high demand, substitute the quantity and rate:

$$R(A_1, S_1) = \begin{array}{l} 2 \times 10 \\ = 20 \end{array} \quad \text{(thousand USD)}.$$

The units are batches times thousand USD per batch, giving thousand USD. The basic plan's rate stays at 10 across its row; under low demand its revenue is $1 \times 10 = 10$ thousand USD because quantity changes. Dependence tells us which input determines a component. It does not require different inputs to have different values: two plans could happen to have equal rates.

A cost rate determined by the plan

Now add a cost assumption: each basic-plan batch uses 3 thousand USD of resources, while each expanded-plan batch uses 4 thousand USD. These are costs *per batch*. Write $c(A_1) = 3$, read "cost rate for alternative one," and $c(A_2) = 4$, in thousand USD per batch. The general plan-based cost rate is $c(A_i)$. If these are the only costs,

$$C(A_i, S_j) = q(A_i, S_j) c(A_i).$$

Keep the revenue and cost calculations separate before finding profit:

$$R(A_1, S_1) = \begin{array}{l} 2 \times 10 \\ = 20 \end{array}, \quad C(A_1, S_1) = \begin{array}{l} 2 \times 3 \\ = 6 \end{array}, \quad P(A_1, S_1) = \begin{array}{l} 20 - 6 \\ = 14 \end{array}.$$

All three results are in thousand USD. Lowercase r and c name rates per batch; uppercase R and C name monetary totals.

Check. Why is $20 - 3$ not the profit for two basic-plan batches? *Suggested answer:* 20 is total revenue, but 3 is cost per batch. First find total cost, $2 \times 3 = 6$, then subtract $20 - 6 = 14$.

A revenue rate determined by demand

Consider a separate pricing variation first: suppose payment comes only from a demand-based rate, with no plan-based payment. A community booking scheme pays 5 thousand USD per batch under high demand and 3 under low demand. Write $r(S_1) = 5$ and $r(S_2) = 3$ thousand USD per batch; read $r(S_1)$ as "revenue rate for state one." The rate $r(S_j)$ is determined by the demand state. For this state-only pricing variation,

$$R(A_i, S_j) = q(A_i, S_j) r(S_j), \qquad R(A_1, S_1) = \begin{array}{l} 2 \times 5 \\ = 10 \end{array}.$$

The result is in thousand USD. The rate is the same down a state column, but total revenue need not be: three batches under A_2 earn $3 \times 5 = 15$ thousand USD in that same state.

Return now to the plan-based pricing version and *extend* it: assume the booking scheme's payment is an addition to the plan payment, so both rates apply to each batch. Then

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)r(A_i) + q(A_i, S_j)r(S_j) \\ &= q(A_i, S_j)[r(A_i) + r(S_j)]. \end{aligned}$$

For the basic plan under high demand, $2 \times 10 + 2 \times 5 = 2(10 + 5) = 30$ thousand USD. Factoring out quantity works because the same two batches earn both rates. We add 10 and 5 because both use thousand USD per batch; multiplying the combined rate by two batches gives money.

A cost rate determined by demand

High-demand bookings may use shared deliveries, while low-demand bookings need separate handling. Suppose handling costs 1 thousand USD per batch in high demand and 2 in low demand. Write $c(S_1) = 1$ and $c(S_2) = 2$ thousand USD per batch; read $c(S_1)$ as "cost rate for state one." The general state-based rate is $c(S_j)$. In a simple variation where handling is the only charged cost,

$$C(A_i, S_j) = q(A_i, S_j)c(S_j), \qquad C(A_1, S_1) = \begin{matrix} 2 \times 1 \\ = 2 \end{matrix}.$$

The result is in thousand USD. Now extend the plan-cost version instead: assume handling is charged *in addition* to the plan's resource cost. Then

$$C(A_i, S_j) = q(A_i, S_j)[c(A_i) + c(S_j)], \qquad C(A_1, S_1) = \begin{matrix} 2(3 + 1) \\ = 8 \end{matrix}.$$

The handling rate stays at 1 down the high-demand column, but its total charge changes from $2 \times 1 = 2$ to $3 \times 1 = 3$ thousand USD as quantity changes. A state-based component does not make the entire total state-only when quantity also depends on the alternative.

Check. Under the version with both revenue rates and both cost rates, what is the basic plan's high-demand profit? *Suggested answer:* Revenue is 30, cost is 8, and profit is $30 - 8 = 22$ thousand USD. Equal state rates across plans do not imply equal totals.

Fixed revenue: first a plan payment, then a state payment

Extend the version with both revenue rates: a sponsor now pays 7 thousand USD for choosing the basic plan, or 9 for choosing the expanded plan, regardless of how many batches are completed. We write $f_r(A_1) = 7$ and $f_r(A_2) = 9$ thousand USD. Read $f_r(A_1)$ as "fixed revenue for alternative one." The letter f indicates a fixed amount; the subscript r identifies it as revenue. The plan determines $f_r(A_i)$. For the basic plan under high demand, revenue becomes

$$R(A_1, S_1) = \begin{matrix} 2(10 + 5) + 7 \\ = 37 \end{matrix} \quad \text{(thousand USD)}.$$

The 7 is added once; it is not a payment for each batch.

Add one more assumption: a community grant pays 6 thousand USD if demand is high and 7 if it is low, independently of the number of batches served. Write $f_r(S_1) = 6$ and $f_r(S_2) = 7$ thousand USD. Read $f_r(S_1)$ as "fixed revenue for state one." The state determines $f_r(S_j)$. The high-demand grant alone contributes 6 thousand USD, whether the plan serves two or three batches. With both fixed payments now included,

$$R(A_1, S_1) = \begin{array}{l} 2(10+5)+7+6 \\ \qquad = 43 \end{array} \quad \text{(thousand USD)}.$$

"Fixed" means independent of quantity within the model. A fixed amount can still differ between plans or states. For a compact *fixed-only* variation, suppose there are no per-batch earnings, only the sponsor and grant. Then $R(A_i, S_j) = f_r(A_i) + f_r(S_j)$, and $R(A_1, S_1) = 7 + 6 = 13$ thousand USD; no quantity is needed.

Fixed cost: first setup, then administration

Return to the version with both per-batch cost rates. Extend it by charging setup once: 8 thousand USD for the basic plan and 10 for the expanded plan. Write $f_c(A_1) = 8$ and $f_c(A_2) = 10$ thousand USD. Read $f_c(A_1)$ as "fixed cost for alternative one." The subscript c identifies cost; the plan determines $f_c(A_i)$. For the basic plan under high demand,

$$C(A_1, S_1) = \begin{array}{l} 2(3+1)+8 \\ \qquad = 16 \end{array} \quad \text{(thousand USD)}.$$

Now add a state-specific administration bill: 5 thousand USD under high demand or 3 under low demand, charged once whichever plan is chosen. Write $f_c(S_1) = 5$ and $f_c(S_2) = 3$ thousand USD. Read $f_c(S_1)$ as "fixed cost for state one." The state determines $f_c(S_j)$. The high-demand bill contributes 5 thousand USD, not 2×5. With setup and administration included,

$$C(A_1, S_1) = \begin{array}{l} 2(3+1)+8+5 \\ \qquad = 21 \end{array} \quad \text{(thousand USD)}.$$

Ordinary positive costs are added *inside total cost*. Total cost is then subtracted once from revenue: $43 - 21 = 22$ thousand USD. Making every cost negative and then subtracting it would reverse its effect.

Check. If the basic plan completes one batch instead of two, does its setup charge become 4? *Suggested answer:* No. Setup remains 8 thousand USD; only the per-batch charges change with quantity.

Assemble the complete models

We have now taught every ingredient. When *all* the arrangements just introduced apply together, their complete models are

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

The four additive revenue components are $q(A_i, S_j)r(A_i)$, $q(A_i, S_j)r(S_j)$, $f_r(A_i)$, and $f_r(S_j)$. The four additive cost components are $q(A_i, S_j)c(A_i)$, $q(A_i, S_j)c(S_j)$, $f_c(A_i)$, and $f_c(S_j)$. Each is a monetary amount. Quantity $q(A_i, S_j)$ is a multiplier, not an extra monetary component.

The later sections count components *within* the revenue and cost models. A one-component example can have one revenue component and one cost component. The two models need not use matching types. For example, if the stated arrangements provide only plan-based per-batch revenue and state-based handling cost, then

$$R(A_i, S_j) = q(A_i, S_j)r(A_i), \qquad C(A_i, S_j) = q(A_i, S_j)c(S_j).$$

If instead there is only a plan sponsorship payment and a state administration bill, the fixed-only models are

$$R(A_i, S_j) = f_r(A_i), \qquad C(A_i, S_j) = f_c(S_j).$$

A two-component version might add a fixed plan payment to per-batch plan revenue; a three-component version could add a fixed state grant as well. Use only components included by the stated arrangements. An excluded component can be treated as zero. A necessary input that is merely missing is unknown, not automatically zero.

Keep the meaning of a signed adjustment

Some financial descriptions may explicitly give a reduction. For example, a 2 thousand USD rebate on a 9 thousand USD bill gives total cost $9 + (-2) = 7$, whereas an additional charge of 2 gives $9 + 2 = 11$. A reduction and a further cost have opposite effects. If a stated model actually gives a negative total cost, subtract it with parentheses: revenue 5 and total cost -2 give profit $5 - (-2) = 7$ thousand USD. These signed cases do not change the basic rule: build the stated totals, then subtract cost once.

The Advanced General Problem now brings the familiar steps together. Its community business is a variation of our service enterprise: it uses the same component values, with one completed service batch counted as one "unit." Its input table connects every quantity, rate, and fixed amount to the notation above. Rates are in thousand USD per unit, while fixed amounts and totals are in thousand USD.

1.4 Advanced General Problem (C1–C3)

A community business must choose the basic operating plan A_1 or expanded operating plan A_2 before demand is revealed as high (S_1) or low (S_2). Its demand study indicates that plan A_1 will complete 2 units in state S_1 and 1 in state S_2, whereas plan A_2 will complete 3 and 2 units, respectively. Each completed service batch counts as one unit. Revenue and cost rates are in thousand USD per unit; fixed amounts and profits are in thousand USD. Each variable rate applies to every unit completed.

The plans command different base selling rates because they offer different services: each unit under A_1 earns an alternative-based revenue rate of 10, while each unit under A_2 earns 11. Demand conditions add a market-based revenue rate of 5 per unit in S_1 or 3 per unit in S_2. The signed agreements also provide plan-based fixed revenue of 7 for A_1 and 9 for A_2, while community funding adds state-based fixed revenue of 6 in S_1 and 7 in S_2.

The business's operating budget assigns an alternative-based cost of 3 per unit to A_1 and 4 per unit to A_2, reflecting the resources used by each plan. Demand-related handling adds a further state-based cost of 1 per unit in S_1 and 2 per unit in S_2. Preparing the plans also creates fixed costs of 8 for A_1 and 10 for A_2, and state-specific administration creates fixed costs of 5 in S_1 and 3 in S_2. The business must combine all four revenue components and all four cost components for each pair, then subtract total cost from total revenue.

Questions/Tasks.

C1 Interpret the decision elements.

C3 Build the payoff table.

C1 - Interpreting the decision.

Element	Description	
Alternatives	A_1**:** Basic operating plan	A_2**:** Expanded operating plan
States of nature	S_1**:** High demand	S_2**:** Low demand
Uncertain event	Customer bookings reveal demand after the business chooses its plan	
Consequences	Service delivery earns income and uses resources; profit measures the result in thousand USD	

Use the complete models developed above: calculate revenue and cost independently, then subtract cost from revenue.

Trace one cell before completing the tables. For (A_1, S_1), use quantity 2, revenue rates 10 and 5, and fixed revenue 7 and 6:

$$R(A_1, S_1) = \begin{array}{l} 2(10+5)+7+6 \\ = 43 \end{array}.$$

Independently, use cost rates 3 and 1 and fixed costs 8 and 5:

$$C(A_1, S_1) = \begin{array}{l} 2(3+1)+8+5 \\ = 21 \end{array}.$$

Finally, $P(A_1, S_1) = 43 - 21 = 22$ thousand USD. This is the entry in row A_1, column S_1 of the final profit table. Apply these same three steps to each remaining pair; do not mix a rate or payment from another plan or state. The following paired tables show revenue first, cost second, and profit third.



C3 - Calculating revenue, cost, and payoff.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Operating plan 1.
A_2	—	Operating plan 2.
S_1	—	Demand outcome 1.
S_2	—	Demand outcome 2.
$q(A_1, S_1)$	2 units	Forecast quantity for A_1 under S_1.
$r(A_1)$	10 thousand USD per unit	Base service selling rate for A_1 (rate per unit).
$r(S_1)$	5 thousand USD per unit	Demand-related revenue rate for S_1 (rate per unit).
$f_r(A_1)$	7 thousand USD	Plan agreement payment for A_1 (fixed amount).
$f_r(S_1)$	6 thousand USD	Community funding for S_1 (fixed amount).
$c(A_1)$	3 thousand USD per unit	Plan operating rate for A_1 (rate per unit).
$c(S_1)$	1 thousand USD per unit	Demand handling rate for S_1 (rate per unit).
$f_c(A_1)$	8 thousand USD	Plan preparation cost for A_1 (fixed amount).
$f_c(S_1)$	5 thousand USD	State administration cost for S_1 (fixed amount).
$q(A_1, S_2)$	1 unit	Forecast quantity for A_1 under S_2.
$r(S_2)$	3 thousand USD per unit	Demand-related revenue rate for S_2 (rate per unit).
$f_r(S_2)$	7 thousand USD	Community funding for S_2 (fixed amount).
$c(S_2)$	2 thousand USD per unit	Demand handling rate for S_2 (rate per unit).
$f_c(S_2)$	3 thousand USD	State administration cost for S_2 (fixed amount).
$q(A_2, S_1)$	3 units	Forecast quantity for A_2 under S_1.
$r(A_2)$	11 thousand USD per unit	Base service selling rate for A_2 (rate per unit).
$f_r(A_2)$	9 thousand USD	Plan agreement payment for A_2 (fixed amount).
$c(A_2)$	4 thousand USD per unit	Plan operating rate for A_2 (rate per unit).
$f_c(A_2)$	10 thousand USD	Plan preparation cost for A_2 (fixed amount).
$q(A_2, S_2)$	2 units	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$2\times$ $[10+5]$ $+7+6$ $=43$	$1\times$ $[10+3]$ $+7+7$ $=27$
A_2	$3\times$ $[11+5]$ $+9+6$ $=63$	$2\times$ $[11+3]$ $+9+7$ $=44$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$2\times$ $[3+1]$ $+8+5$ $=21$	$1\times$ $[3+2]$ $+8+3$ $=16$
A_2	$3\times$ $[4+1]$ $+10+5$ $=30$	$2\times$ $[4+2]$ $+10+3$ $=25$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$43-21$ $=22$	$27-16$ $=11$
A_2	$63-30$ $=33$	$44-25$ $=19$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	22	11
A_2	33	19



1.5 Reference and classroom solving routine

Notation already used. Monetary units below are thousand USD; "per unit" refers to the quantity unit stated in the problem.

Symbol	Meaning and units
A_i, S_j	Alternative and state labels; i and j are indices, not money or ranks.
$q(A_i, S_j)$	Quantity for that pair, in batches or another stated quantity unit.
$r(A_i)$, $r(S_j)$	Revenue rates determined by the plan or the state, in thousand USD per unit.
$c(A_i)$, $c(S_j)$	Cost rates determined by the plan or the state, in thousand USD per unit.
$f_r(A_i)$, $f_r(S_j)$	Fixed revenue determined by the plan or the state, in thousand USD; add once.
$f_c(A_i)$, $f_c(S_j)$	Fixed cost determined by the plan or the state, in thousand USD; add once inside cost.
$R(A_i, S_j)$	Total revenue for the pair, in thousand USD.
$C(A_i, S_j)$	Total cost for the pair, in thousand USD.
$P(A_i, S_j)$	Profit/payoff for the pair: revenue minus cost, in thousand USD.

Practical solving procedure.

1. Identify who chooses, the alternatives, the uncertain process, its states, the timing, and the consequences in their economic context (C1).
2. Map choices and states to symbols. List all alternative–state pairs.
3. Record inputs in a *Symbol | Value | Meaning in this problem* table, with units. Distinguish direct totals, quantities, rates, and fixed amounts; check that necessary inputs are supplied.
4. Calculate revenue for each pair from its stated components, or record the directly supplied total. Match quantity units to rate units.
5. Calculate cost independently for the same pair, or record the supplied total. Add each applicable fixed amount once.
6. Subtract total cost from total revenue (C3). Match the symbolic calculation to the full numerical substitution and its result.
7. Construct the profit payoff table (C3): alternatives in rows, states in columns, profits only in cells, and monetary units stated. Keep the same row and column order throughout.

Final checks with suggested answers.

- Is 10 thousand USD per batch a total revenue? *No: it is a rate. Two batches earn 20 thousand USD at that rate.*
- Does a fixed payment of 7 thousand USD become 14 for two batches? *No: add it once. Only per-batch amounts are multiplied by quantity.*
- With revenue 43 and cost 21 thousand USD, what enters the payoff table? *Profit:* $43 - 21 = 22$ *thousand USD.*



2 Direct Revenue and Cost Models

Problems:

2.1. Local Café Investment (C1–C3)
2.2. Green Energy Production Mix (C1–C3)

2.1 Local Café Investment (C1–C3)

A small café must decide whether to launch a line of artisan desserts or keep its current menu before learning whether daily customer traffic will be high or low. The owner prepared direct total-revenue forecasts from expected café sales and direct total-cost budgets covering ingredients, staffing, preparation, and other resources. Every amount and the resulting daily profit are in thousand USD.

Launching the dessert line could generate revenue of 58 when traffic is high, when more customers buy the new products, and 27 when traffic is low. The matching operating budgets are 16 and 19; the low-traffic cost remains comparatively high because ingredients and preparation must be arranged before actual traffic is known. Keeping the current menu instead is forecast to produce revenue of 49 under high traffic and 41 under low traffic, with established-menu operating costs of 21 in either state. The owner must calculate profit for all four outcomes.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Launch dessert line	A_2: Keep current menu
States of nature	S_1: High traffic	S_2: Low traffic
Uncertain event	Actual foot traffic after the decision	
Consequences	Profit in thousand USD for each alternative–state pair	



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Launch dessert line.
A_2	—	Keep current menu.
S_1	—	High traffic.
S_2	—	Low traffic.

Symbol	Value	Meaning in this problem.
$R(A_1, S_1)$	58 thousand USD	Forecast total revenue for A_1 under S_1.
$C(A_1, S_1)$	16 thousand USD	Budgeted total cost for A_1 under S_1.
$R(A_1, S_2)$	27 thousand USD	Forecast total revenue for A_1 under S_2.
$C(A_1, S_2)$	19 thousand USD	Budgeted total cost for A_1 under S_2.
$R(A_2, S_1)$	49 thousand USD	Forecast total revenue for A_2 under S_1.
$C(A_2, S_1)$	21 thousand USD	Budgeted total cost for A_2 under S_1.
$R(A_2, S_2)$	41 thousand USD	Forecast total revenue for A_2 under S_2.
$C(A_2, S_2)$	21 thousand USD	Budgeted total cost for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$58 - 16$ $= 42$	$27 - 19$ $= 8$
A_2	$49 - 21$ $= 28$	$41 - 21$ $= 20$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	42	8
A_2	28	20



2.2 Green Energy Production Mix (C1–C3)

A renewable-energy company must build wind turbines, solar farms, or a balanced hybrid system before the year's weather is known. Annual electricity sales and the operating requirements of each generation system have been forecast directly for windy, sunny, and cloudy conditions. All revenue, cost, and profit figures are in thousand USD.

Wind turbines are expected to earn revenue of 132 in windy weather, 78 in sunny weather, and 62 in cloudy weather because their electricity output changes with conditions; the corresponding operating and maintenance costs are 42, 33, and 32. Solar farms instead are forecast to earn 77, 143, and 63 in the same respective states, with costs of 42, 48, and 38, reflecting the resources needed at each output level.

The hybrid system spreads production across both sources, producing forecast revenue of 112 when it is windy, 107 when it is sunny, and 96 when it is cloudy. Its corresponding operating costs are 42, 42, and 36. Management must subtract the appropriate direct cost from revenue for each strategy–weather combination.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Wind turbines	A_2: Solar farms	A_3: Hybrid system
States of nature	S_1: Windy	S_2: Sunny	S_3: Cloudy
Uncertain event	Weather outcome affecting electricity generation		
Consequences	Profit in thousand USD for each strategy–weather combination		



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Wind turbines.
A_2	—	Solar farms.
A_3	—	Hybrid system.
S_1	—	Windy.
S_2	—	Sunny.
S_3	—	Cloudy.
$R(A_1, S_1)$	132 thousand USD	Forecast total revenue for A_1 under S_1.
$C(A_1, S_1)$	42 thousand USD	Budgeted total cost for A_1 under S_1.
$R(A_1, S_2)$	78 thousand USD	Forecast total revenue for A_1 under S_2.
$C(A_1, S_2)$	33 thousand USD	Budgeted total cost for A_1 under S_2.
$R(A_1, S_3)$	62 thousand USD	Forecast total revenue for A_1 under S_3.
$C(A_1, S_3)$	32 thousand USD	Budgeted total cost for A_1 under S_3.
$R(A_2, S_1)$	77 thousand USD	Forecast total revenue for A_2 under S_1.
$C(A_2, S_1)$	42 thousand USD	Budgeted total cost for A_2 under S_1.
$R(A_2, S_2)$	143 thousand USD	Forecast total revenue for A_2 under S_2.
$C(A_2, S_2)$	48 thousand USD	Budgeted total cost for A_2 under S_2.
$R(A_2, S_3)$	63 thousand USD	Forecast total revenue for A_2 under S_3.
$C(A_2, S_3)$	38 thousand USD	Budgeted total cost for A_2 under S_3.
$R(A_3, S_1)$	112 thousand USD	Forecast total revenue for A_3 under S_1.
$C(A_3, S_1)$	42 thousand USD	Budgeted total cost for A_3 under S_1.
$R(A_3, S_2)$	107 thousand USD	Forecast total revenue for A_3 under S_2.
$C(A_3, S_2)$	42 thousand USD	Budgeted total cost for A_3 under S_2.
$R(A_3, S_3)$	96 thousand USD	Forecast total revenue for A_3 under S_3.
$C(A_3, S_3)$	36 thousand USD	Budgeted total cost for A_3 under S_3.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Profit

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$	$R(A_1,S_3)-C(A_1,S_3)$ $=P(A_1,S_3)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$	$R(A_2,S_3)-C(A_2,S_3)$ $=P(A_2,S_3)$
A_3	$R(A_3,S_1)-C(A_3,S_1)$ $=P(A_3,S_1)$	$R(A_3,S_2)-C(A_3,S_2)$ $=P(A_3,S_2)$	$R(A_3,S_3)-C(A_3,S_3)$ $=P(A_3,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$132-42$ $=90$	$78-33$ $=45$	$62-32$ $=30$
A_2	$77-42$ $=35$	$143-48$ $=95$	$63-38$ $=25$
A_3	$112-42$ $=70$	$107-42$ $=65$	$96-36$ $=60$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2	S_3
A_1	90	45	30
A_2	35	95	25
A_3	70	65	60



3 One-Component Models

Problems:

3.1. Custom Printing Product Line (C1–C3)

3.2. Event Equipment Fleet (C1–C3)

3.3. Community Arts Sponsorship (C1–C3)

3.4. Distribution Contract Fulfilment (C1–C3)

3.1 Custom Printing Product Line (C1–C3)

A custom-printing cooperative must choose to produce premium notebooks or canvas bags before it knows whether the market will bring strong demand with stable material prices or weak demand with higher material prices. Orders already under discussion indicate that premium notebooks will sell 6 production lots in the strong state and 4 in the weak state, while canvas bags will sell 5 and 3 lots, respectively.

Because the two products have different selling prices, each notebook lot earns an alternative-based revenue rate of 12 thousand USD and each canvas-bag lot earns 15 thousand USD. Suppliers quote by market condition rather than product: stable prices make the cost 5 thousand USD per lot in the strong state, while higher prices raise it to 8 thousand USD per lot in the weak state. These are the only revenue and cost components, so the cooperative must apply both rates to the appropriate forecast lot quantity.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Produce premium notebooks	A_2**:** Produce canvas bags
States of nature	S_1**:** Strong demand and stable material prices	S_2**:** Weak demand and higher material prices
Uncertain event	Market conditions revealed after the product-line decision	
Consequences	Profit in thousand USD for each product-line–market-state pair	

The only revenue component is an alternative-based variable component:

$$R(A_i, S_j) = q(A_i, S_j)r(A_i).$$

The only cost component is a state-based variable component:

$$C(A_i, S_j) = q(A_i, S_j)c(S_j).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j)r(A_i) - q(A_i, S_j)c(S_j).$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Produce premium notebooks.
A_2	—	Produce canvas bags.
S_1	—	Strong demand and stable material prices.
S_2	—	Weak demand and higher material prices.
$q(A_1, S_1)$	6 production lots	Forecast quantity for A_1 under S_1.
$r(A_1)$	12 thousand USD per production lot	Product selling rate for A_1 (rate per unit).
$c(S_1)$	5 thousand USD per production lot	Material price per lot for S_1 (rate per unit).
$q(A_1, S_2)$	4 production lots	Forecast quantity for A_1 under S_2.
$c(S_2)$	8 thousand USD per production lot	Material price per lot for S_2 (rate per unit).
$q(A_2, S_1)$	5 production lots	Forecast quantity for A_2 under S_1.
$r(A_2)$	15 thousand USD per production lot	Product selling rate for A_2 (rate per unit).
$q(A_2, S_2)$	3 production lots	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times r(A_1)$ $= R(A_1, S_1)$	$q(A_1, S_2) \times r(A_1)$ $= R(A_1, S_2)$
A_2	$q(A_2, S_1) \times r(A_2)$ $= R(A_2, S_1)$	$q(A_2, S_2) \times r(A_2)$ $= R(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×12 $= 72$	4×12 $= 48$
A_2	5×15 $= 75$	3×15 $= 45$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×5 $=30$	4×8 $=32$
A_2	5×5 $=25$	3×8 $=24$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$72-30$ $=42$	$48-32$ $=16$
A_2	$75-25$ $=50$	$45-24$ $=21$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	42	16
A_2	50	21



3.2 Event Equipment Fleet (C1–C3)

An event-equipment provider must prepare either a compact fleet or an expanded fleet before rental demand is known. Booking enquiries indicate that the compact fleet will fulfil 6 rental contracts under high demand and 3 under low demand; the expanded fleet has capacity for 8 and 4 contracts in those respective states.

Market prices make every completed contract earn 9 thousand USD when demand is high and 6 thousand USD when it is low, regardless of the fleet selected. Preparing, transporting, and inspecting the equipment requires a fixed outlay of 22 thousand USD for the compact fleet or 34 thousand USD for the expanded fleet. No other revenue or cost component is modeled, so the provider must compare state-based contract revenue with the chosen fleet's fixed preparation cost.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Prepare a compact fleet	A_2: Prepare an expanded fleet
States of nature	S_1: High rental demand	S_2: Low rental demand
Uncertain event	Rental demand revealed after the fleet-size decision	
Consequences	Profit in thousand USD for each fleet–demand pair	

The only revenue component is a state-based variable component:

$$R(A_i, S_j) = q(A_i, S_j)r(S_j).$$

The only cost component is an alternative-based fixed component:

$$C(A_i, S_j) = f_c(A_i).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j)r(S_j) - f_c(A_i).$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Prepare a compact fleet.
A_2	—	Prepare an expanded fleet.
S_1	—	High rental demand.
S_2	—	Low rental demand.
$q(A_1, S_1)$	6 rental contracts	Forecast quantity for A_1 under S_1.
$r(S_1)$	9 thousand USD per rental contract	Market rental rate for S_1 (rate per unit).
$f_c(A_1)$	22 thousand USD	Fleet preparation cost for A_1 (fixed amount).
$q(A_1, S_2)$	3 rental contracts	Forecast quantity for A_1 under S_2.
$r(S_2)$	6 thousand USD per rental contract	Market rental rate for S_2 (rate per unit).
$q(A_2, S_1)$	8 rental contracts	Forecast quantity for A_2 under S_1.
$f_c(A_2)$	34 thousand USD	Fleet preparation cost for A_2 (fixed amount).
$q(A_2, S_2)$	4 rental contracts	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×9 $=54$	3×6 $=18$
A_2	8×9 $=72$	4×6 $=24$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$54-22$ $=32$	$18-22$ $=-4$
A_2	$72-34$ $=38$	$24-34$ $=-10$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	32	-4
A_2	38	-10

The negative payoffs in the low-demand state represent losses because the fixed preparation cost is greater than the revenue earned.



3.3 Community Arts Sponsorship (C1–C3)

A community arts organisation must accept either a standard or premium sponsorship agreement before officials confirm whether its event faces routine permit requirements or enhanced safety requirements. The sponsor guarantees a fixed payment of 52 thousand USD under the standard agreement or 68 thousand USD under the premium agreement, so this revenue depends only on the chosen agreement.

The authorities' requirements determine the event's fixed regulatory and safety bill: routine permits cost 18 thousand USD, while enhanced safety measures cost 31 thousand USD, whichever sponsorship agreement was selected. Because neither payment changes with attendance or output, no quantity is required; the organisation simply subtracts the relevant state-based fixed cost from its agreement-based fixed revenue.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Accept the standard sponsorship agreement	A_2: Accept the premium sponsorship agreement
States of nature	S_1: Routine permit requirements	S_2: Enhanced safety requirements
Uncertain event	Event requirements confirmed after the sponsorship decision	
Consequences	Profit in thousand USD for each sponsorship–requirement pair	

The only revenue component is an alternative-based fixed component:

$$R(A_i, S_j) = f_r(A_i).$$

The only cost component is a state-based fixed component:

$$C(A_i, S_j) = f_c(S_j).$$

Therefore,

$$P(A_i, S_j) = f_r(A_i) - f_c(S_j).$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Accept the standard sponsorship agreement.
A_2	—	Accept the premium sponsorship agreement.
S_1	—	Routine permit requirements.
S_2	—	Enhanced safety requirements.
$f_r(A_1)$	52 thousand USD	Sponsorship payment for A_1 (fixed amount).
$f_c(S_1)$	18 thousand USD	Regulatory and safety bill for S_1 (fixed amount).
$f_c(S_2)$	31 thousand USD	Regulatory and safety bill for S_2 (fixed amount).
$f_r(A_2)$	68 thousand USD	Sponsorship payment for A_2 (fixed amount).

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$52 - 18$ $= 34$	$52 - 31$ $= 21$
A_2	$68 - 18$ $= 50$	$68 - 31$ $= 37$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	34	21
A_2	50	37



3.4 Distribution Contract Fulfilment (C1–C3)

A distribution company must choose manual or automated packing before a new contract's volume is confirmed as standard or high. Capacity forecasts show manual packing processing 4 hundreds of orders under the standard-volume contract and 7 under the high-volume contract, while automated packing processes 5 and 8 hundreds, respectively.

The customer pays a fixed amount for the realised contract size: 70 thousand USD for standard volume or 104 thousand USD for high volume, regardless of the fulfilment method. Labour makes manual processing cost 7 thousand USD per hundred orders, whereas automation lowers that method-based rate to 5 thousand USD per hundred orders. With no other components, the company must deduct the chosen method's quantity-based processing cost from the state's fixed contract payment.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Use manual packing	A_2**:** Use automated packing
States of nature	S_1**:** Standard-volume contract	S_2**:** High-volume contract
Uncertain event	Contract volume revealed after the fulfilment-method decision	
Consequences	Profit in thousand USD for each fulfilment-method–volume pair	

The only revenue component is a state-based fixed component:

$$R(A_i, S_j) = f_r(S_j).$$

The only cost component is an alternative-based variable component:

$$C(A_i, S_j) = q(A_i, S_j)c(A_i).$$

Therefore,

$$P(A_i, S_j) = f_r(S_j) - q(A_i, S_j)c(A_i).$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use manual packing.
A_2	—	Use automated packing.
S_1	—	Standard-volume contract.
S_2	—	High-volume contract.
$f_r(S_1)$	70 thousand USD	Contract payment for S_1 (fixed amount).
$q(A_1, S_1)$	4 hundreds of orders	Forecast quantity for A_1 under S_1.
$c(A_1)$	7 thousand USD per hundred orders	Packing labour rate for A_1 (rate per unit).
$f_r(S_2)$	104 thousand USD	Contract payment for S_2 (fixed amount).
$q(A_1, S_2)$	7 hundreds of orders	Forecast quantity for A_1 under S_2.
$q(A_2, S_1)$	5 hundreds of orders	Forecast quantity for A_2 under S_1.
$c(A_2)$	5 thousand USD per hundred orders	Packing labour rate for A_2 (rate per unit).
$q(A_2, S_2)$	8 hundreds of orders	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times c(A_1)$ $= C(A_1, S_1)$	$q(A_1, S_2) \times c(A_1)$ $= C(A_1, S_2)$
A_2	$q(A_2, S_1) \times c(A_2)$ $= C(A_2, S_1)$	$q(A_2, S_2) \times c(A_2)$ $= C(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	4×7 $= 28$	7×7 $= 49$
A_2	5×5 $= 25$	8×5 $= 40$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1, S_1) - C(A_1, S_1)$ $= P(A_1, S_1)$	$R(A_1, S_2) - C(A_1, S_2)$ $= P(A_1, S_2)$
A_2	$R(A_2, S_1) - C(A_2, S_1)$ $= P(A_2, S_1)$	$R(A_2, S_2) - C(A_2, S_2)$ $= P(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$70 - 28$ $= 42$	$104 - 49$ $= 55$
A_2	$70 - 25$ $= 45$	$104 - 40$ $= 64$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	42	55
A_2	45	64



4 Two-Component Models

Problems:

4.1. Farm Produce Subscription Plans (C1–C3)

4.2. Mobile Catering Configuration (C1–C3)

4.3. Online Training Programme (C1–C3)

4.4. Community Venue Membership (C1–C3)

4.5. Urban Delivery Fleet (C1–C3)

4.6. Data-Service Contract (C1–C3)

4.1 Farm Produce Subscription Plans (C1–C3)

A farm cooperative must offer either a standard or premium produce box before seasonal demand is known to be strong or weak. Its subscription forecast gives the standard box 7 hundreds of subscriptions in the strong state and 4 in the weak state, and the premium box 6 and 3 hundreds, respectively.

The standard box's contents generate a plan-based revenue rate of 8 thousand USD per hundred subscriptions, while the premium contents generate 11. Strong market conditions add 3 thousand USD per hundred subscriptions to either plan's revenue; weak conditions add 1. Packing inputs cost 4 thousand USD per hundred standard subscriptions or 6 per hundred premium subscriptions. Finally, seasonal distribution arrangements create a fixed cost of 9 thousand USD under strong demand and 13 under weak demand. The cooperative must combine the two revenue rates, then deduct both the plan-based variable cost and the state-based fixed distribution cost.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Offer a standard produce box	A_2: Offer a premium produce box
States of nature	S_1: Strong seasonal demand	S_2: Weak seasonal demand
Uncertain event	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Profit in thousand USD for each plan–market-state pair	

Revenue contains alternative-based and state-based variable components:

$$R(A_i, S_j) = q(A_i, S_j)\left[r(A_i) + r(S_j)\right].$$

Cost contains an alternative-based variable component and a state-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)c(A_i) + f_c(S_j).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j)\left[r(A_i) + r(S_j)\right] - \left[q(A_i, S_j)c(A_i) + f_c(S_j)\right].$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Offer a standard produce box.
A_2	—	Offer a premium produce box.
S_1	—	Strong seasonal demand.
S_2	—	Weak seasonal demand.
$q(A_1, S_1)$	7 hundreds of subscriptions	Forecast quantity for A_1 under S_1.
$r(A_1)$	8 thousand USD per hundred subscriptions	Produce-box selling rate for A_1 (rate per unit).
$r(S_1)$	3 thousand USD per hundred subscriptions	Seasonal market supplement for S_1 (rate per unit).
$c(A_1)$	4 thousand USD per hundred subscriptions	Packing input rate for A_1 (rate per unit).
$f_c(S_1)$	9 thousand USD	Seasonal distribution cost for S_1 (fixed amount).
$q(A_1, S_2)$	4 hundreds of subscriptions	Forecast quantity for A_1 under S_2.
$r(S_2)$	1 thousand USD per hundred subscriptions	Seasonal market supplement for S_2 (rate per unit).
$f_c(S_2)$	13 thousand USD	Seasonal distribution cost for S_2 (fixed amount).
$q(A_2, S_1)$	6 hundreds of subscriptions	Forecast quantity for A_2 under S_1.
$r(A_2)$	11 thousand USD per hundred subscriptions	Produce-box selling rate for A_2 (rate per unit).
$c(A_2)$	6 thousand USD per hundred subscriptions	Packing input rate for A_2 (rate per unit).
$q(A_2, S_2)$	3 hundreds of subscriptions	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7\times$ $[8+3]$ $=77$	$4\times$ $[8+1]$ $=36$
A_2	$6\times$ $[11+3]$ $=84$	$3\times$ $[11+1]$ $=36$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(A_1)$ $+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times c(A_1)$ $+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(A_2)$ $+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times c(A_2)$ $+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	7×4 $+9$ $=37$	4×4 $+13$ $=29$
A_2	6×6 $+9$ $=45$	3×6 $+13$ $=31$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$77-37$ $=40$	$36-29$ $=7$
A_2	$84-45$ $=39$	$36-31$ $=5$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	40	7
A_2	39	5



4.2 Mobile Catering Configuration (C1–C3)

A mobile-catering company must prepare a compact catering station or a full mobile kitchen before learning whether the event will be busy with stable ingredient prices or quiet with higher ingredient prices. Booking forecasts indicate 8 hundreds of service orders for the compact station in the busy state and 4 in the quiet state, compared with 10 and 5 hundreds for the full kitchen.

Menu sales earn a configuration-based 7 thousand USD per hundred orders from the compact station or 9 from the full kitchen. Booking agreements add fixed revenue of 18 thousand USD for the compact configuration and 25 for the full configuration. Ingredients cost 3 thousand USD per hundred orders when prices are stable and 5 when prices are higher. In addition, setting up the compact station costs a fixed 20 thousand USD, while preparing the full kitchen costs 32. The company must retain each component's alternative- or state-based role when calculating profit.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Prepare a compact catering station	A_2**:** Prepare a full mobile kitchen
States of nature	S_1**:** Busy event with stable ingredient prices	S_2**:** Quiet event with higher ingredient prices
Uncertain event	Event demand and ingredient-market conditions revealed after the configuration decision	
Consequences	Profit in thousand USD for each configuration–market-state pair	

Revenue contains alternative-based variable and fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r(A_i) + f_r(A_i).$$

Cost contains a state-based variable component and an alternative-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)c(S_j) + f_c(A_i).$$

Therefore,

$$P(A_i, S_j) = [q(A_i, S_j)r(A_i) + f_r(A_i)] - [q(A_i, S_j)c(S_j) + f_c(A_i)]\,.$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Prepare a compact catering station.
A_2	—	Prepare a full mobile kitchen.
S_1	—	Busy event with stable ingredient prices.
S_2	—	Quiet event with higher ingredient prices.
$q(A_1, S_1)$	8 hundreds of service orders	Forecast quantity for A_1 under S_1.
$r(A_1)$	7 thousand USD per hundred service orders	Menu sales rate for A_1 (rate per unit).
$f_r(A_1)$	18 thousand USD	Booking agreement payment for A_1 (fixed amount).
$c(S_1)$	3 thousand USD per hundred service orders	Ingredient rate for S_1 (rate per unit).
$f_c(A_1)$	20 thousand USD	Configuration setup cost for A_1 (fixed amount).
$q(A_1, S_2)$	4 hundreds of service orders	Forecast quantity for A_1 under S_2.
$c(S_2)$	5 thousand USD per hundred service orders	Ingredient rate for S_2 (rate per unit).
$q(A_2, S_1)$	10 hundreds of service orders	Forecast quantity for A_2 under S_1.
$r(A_2)$	9 thousand USD per hundred service orders	Menu sales rate for A_2 (rate per unit).
$f_r(A_2)$	25 thousand USD	Booking agreement payment for A_2 (fixed amount).
$f_c(A_2)$	32 thousand USD	Configuration setup cost for A_2 (fixed amount).
$q(A_2, S_2)$	5 hundreds of service orders	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(A_1)$ $+ f_r(A_1)$ $= R(A_1,S_1)$	$q(A_1,S_2)\times r(A_1)$ $+ f_r(A_1)$ $= R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(A_2)$ $+ f_r(A_2)$ $= R(A_2,S_1)$	$q(A_2,S_2)\times r(A_2)$ $+ f_r(A_2)$ $= R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	8×7 $+18$ $=74$	4×7 $+18$ $=46$
A_2	10×9 $+25$ $=115$	5×9 $+25$ $=70$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(S_1)$ $+ f_c(A_1)$ $= C(A_1,S_1)$	$q(A_1,S_2)\times c(S_2)$ $+ f_c(A_1)$ $= C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(S_1)$ $+ f_c(A_2)$ $= C(A_2,S_1)$	$q(A_2,S_2)\times c(S_2)$ $+ f_c(A_2)$ $= C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	8×3 $+20$ $=44$	4×5 $+20$ $=40$
A_2	10×3 $+32$ $=62$	5×5 $+32$ $=57$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $= P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $= P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $= P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $= P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$74-44$ $=30$	$46-40$ $=6$
A_2	$115-62$ $=53$	$70-57$ $=13$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	30	6
A_2	53	13

4.3 Online Training Programme (C1–C3)

An online-training provider must offer its foundation or advanced programme before enrolment is known to come from corporate clients or individuals. Past enquiries imply 5 enrolment groups for the foundation programme under corporate enrolment and 3 under individual enrolment; the advanced programme expects 4 and 2 groups, respectively.

Tuition generates 14 thousand USD per foundation group or 20 thousand USD per advanced group. The enrolment channel also provides a fixed payment of 12 thousand USD for corporate enrolment or 7 for individual enrolment. Delivering a group costs 6 thousand USD in the corporate state and 8 in the individual state, while state-specific platform arrangements add fixed costs of 10 and 14, respectively. The provider must apply both state-based costs to the forecast groups while keeping the fixed amounts independent of quantity.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Offer the foundation programme	A_2**:** Offer the advanced programme
States of nature	S_1**:** Corporate enrolment	S_2**:** Individual enrolment
Uncertain event	Enrolment type revealed after the programme decision	
Consequences	Profit in thousand USD for each programme–enrolment pair	

Revenue contains an alternative-based variable component and a state-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)r(A_i) + f_r(S_j).$$

Cost contains state-based variable and fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c(S_j) + f_c(S_j).$$

Therefore,

$$P(A_i, S_j) = [q(A_i, S_j)r(A_i) + f_r(S_j)] - [q(A_i, S_j)c(S_j) + f_c(S_j)]\,.$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Offer the foundation programme.
A_2	—	Offer the advanced programme.
S_1	—	Corporate enrolment.
S_2	—	Individual enrolment.
$q(A_1, S_1)$	5 enrolment groups	Forecast quantity for A_1 under S_1.
$r(A_1)$	14 thousand USD per enrolment group	Tuition rate for A_1 (rate per unit).
$f_r(S_1)$	12 thousand USD	Enrolment-channel payment for S_1 (fixed amount).
$c(S_1)$	6 thousand USD per enrolment group	Group delivery rate for S_1 (rate per unit).
$f_c(S_1)$	10 thousand USD	Platform arrangement cost for S_1 (fixed amount).
$q(A_1, S_2)$	3 enrolment groups	Forecast quantity for A_1 under S_2.
$f_r(S_2)$	7 thousand USD	Enrolment-channel payment for S_2 (fixed amount).
$c(S_2)$	8 thousand USD per enrolment group	Group delivery rate for S_2 (rate per unit).
$f_c(S_2)$	14 thousand USD	Platform arrangement cost for S_2 (fixed amount).
$q(A_2, S_1)$	4 enrolment groups	Forecast quantity for A_2 under S_1.
$r(A_2)$	20 thousand USD per enrolment group	Tuition rate for A_2 (rate per unit).
$q(A_2, S_2)$	2 enrolment groups	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(A_1)$ $+ f_r(S_1)$ $= R(A_1,S_1)$	$q(A_1,S_2)\times r(A_1)$ $+ f_r(S_2)$ $= R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(A_2)$ $+ f_r(S_1)$ $= R(A_2,S_1)$	$q(A_2,S_2)\times r(A_2)$ $+ f_r(S_2)$ $= R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	5×14 $+12$ $=82$	3×14 $+7$ $=49$
A_2	4×20 $+12$ $=92$	2×20 $+7$ $=47$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(S_1)$ $+ f_c(S_1)$ $= C(A_1,S_1)$	$q(A_1,S_2)\times c(S_2)$ $+ f_c(S_2)$ $= C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(S_1)$ $+ f_c(S_1)$ $= C(A_2,S_1)$	$q(A_2,S_2)\times c(S_2)$ $+ f_c(S_2)$ $= C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	5×6 $+10$ $=40$	3×8 $+14$ $=38$
A_2	4×6 $+10$ $=34$	2×8 $+14$ $=30$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $= P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $= P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $= P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $= P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$82-40$ $=42$	$49-38$ $=11$
A_2	$92-34$ $=58$	$47-30$ $=17$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	42	11
A_2	58	17

4.4 Community Venue Membership (C1–C3)

A community organisation must select standard or expanded venue membership before booking demand is known to be high or low. Reservations are forecast at 6 bookings for standard membership under high demand and 3 under low demand, while expanded membership enables 8 and 4 bookings, respectively.

The market supports revenue of 10 thousand USD per booking in the high-demand state and 7 in the low-demand state. A sponsor also contributes a fixed 15 thousand USD when the standard plan is selected or 22 for the expanded plan. Annual membership itself costs a fixed 28 thousand USD for the standard plan and 40 for the expanded plan, while permits add a state-based fixed cost of 8 under high demand or 13 under low demand. The organisation must calculate revenue and subtract both fixed cost sources without multiplying them by bookings.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Select the standard venue membership	A_2**:** Select the expanded venue membership
States of nature	S_1**:** High booking demand	S_2**:** Low booking demand
Uncertain event	Booking demand revealed after the membership-plan decision	
Consequences	Profit in thousand USD for each membership–demand pair	

Revenue contains a state-based variable component and an alternative-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)r(S_j) + f_r(A_i).$$

Cost contains alternative-based and state-based fixed components:

$$C(A_i, S_j) = f_c(A_i) + f_c(S_j).$$

Therefore,

$$P(A_i, S_j) = [q(A_i, S_j)r(S_j) + f_r(A_i)] - [f_c(A_i) + f_c(S_j)]\,.$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Select the standard venue membership.
A_2	—	Select the expanded venue membership.
S_1	—	High booking demand.
S_2	—	Low booking demand.
$q(A_1, S_1)$	6 bookings	Forecast quantity for A_1 under S_1.
$r(S_1)$	10 thousand USD per booking	Booking revenue rate for S_1 (rate per unit).
$f_r(A_1)$	15 thousand USD	Sponsor contribution for A_1 (fixed amount).
$f_c(A_1)$	28 thousand USD	Annual membership cost for A_1 (fixed amount).
$f_c(S_1)$	8 thousand USD	Permit cost for S_1 (fixed amount).
$q(A_1, S_2)$	3 bookings	Forecast quantity for A_1 under S_2.
$r(S_2)$	7 thousand USD per booking	Booking revenue rate for S_2 (rate per unit).
$f_c(S_2)$	13 thousand USD	Permit cost for S_2 (fixed amount).
$q(A_2, S_1)$	8 bookings	Forecast quantity for A_2 under S_1.
$f_r(A_2)$	22 thousand USD	Sponsor contribution for A_2 (fixed amount).
$f_c(A_2)$	40 thousand USD	Annual membership cost for A_2 (fixed amount).
$q(A_2, S_2)$	4 bookings	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(S_1)$ $+f_r(A_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times r(S_2)$ $+f_r(A_1)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(S_1)$ $+f_r(A_2)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times r(S_2)$ $+f_r(A_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×10 $+15$ $=75$	3×7 $+15$ $=36$
A_2	8×10 $+22$ $=102$	4×7 $+22$ $=50$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$28+8$ $=36$	$28+13$ $=41$
A_2	$40+8$ $=48$	$40+13$ $=53$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$75-36$ $=39$	$36-41$ $=-5$
A_2	$102-48$ $=54$	$50-53$ $=-3$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	39	-5
A_2	54	-3

The negative payoffs under low booking demand represent losses because the combined fixed costs exceed the revenue earned.



4.5 Urban Delivery Fleet (C1–C3)

An urban-delivery company must deploy conventional vans or electric cargo bikes before conditions resolve into high demand with light traffic or low demand with heavy traffic. Route forecasts assign the vans 9 delivery batches in the first state and 5 in the second, while cargo bikes can complete 12 and 6 batches, respectively.

Customer contracts pay 8 thousand USD per batch in the high-demand state and 6 in the low-demand state. Meeting state targets also brings a fixed performance payment of 14 thousand USD with high demand or 8 with low demand. Van operation costs 3 thousand USD per batch and cargo-bike operation 2; traffic then adds 1 thousand USD per batch in light traffic or 3 in heavy traffic. The company must combine the fleet-based and traffic-based rates for every delivered batch before deducting costs.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Use a conventional van fleet	A_2**:** Use an electric cargo-bike fleet
States of nature	S_1**:** High demand with light traffic	S_2**:** Low demand with heavy traffic
Uncertain event	Delivery demand and traffic conditions revealed after the fleet decision	
Consequences	Profit in thousand USD for each fleet–market-state pair	

Revenue contains state-based variable and fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r(S_j) + f_r(S_j).$$

Cost contains alternative-based and state-based variable components:

$$C(A_i, S_j) = q(A_i, S_j)\left[c(A_i) + c(S_j)\right].$$

Therefore,

$$P(A_i, S_j) = \left[q(A_i, S_j)r(S_j) + f_r(S_j)\right] - q(A_i, S_j)\left[c(A_i) + c(S_j)\right].$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use a conventional van fleet.
A_2	—	Use an electric cargo-bike fleet.
S_1	—	High demand with light traffic.
S_2	—	Low demand with heavy traffic.
$q(A_1, S_1)$	9 delivery batches	Forecast quantity for A_1 under S_1.
$r(S_1)$	8 thousand USD per delivery batch	Customer contract rate for S_1 (rate per unit).
$f_r(S_1)$	14 thousand USD	Performance payment for S_1 (fixed amount).
$c(A_1)$	3 thousand USD per delivery batch	Fleet operating rate for A_1 (rate per unit).
$c(S_1)$	1 thousand USD per delivery batch	Traffic-related rate for S_1 (rate per unit).
$q(A_1, S_2)$	5 delivery batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	6 thousand USD per delivery batch	Customer contract rate for S_2 (rate per unit).
$f_r(S_2)$	8 thousand USD	Performance payment for S_2 (fixed amount).
$c(S_2)$	3 thousand USD per delivery batch	Traffic-related rate for S_2 (rate per unit).
$q(A_2, S_1)$	12 delivery batches	Forecast quantity for A_2 under S_1.
$c(A_2)$	2 thousand USD per delivery batch	Fleet operating rate for A_2 (rate per unit).
$q(A_2, S_2)$	6 delivery batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(S_1)$ $+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times r(S_2)$ $+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(S_1)$ $+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times r(S_2)$ $+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	9×8 $+14$ $=86$	5×6 $+8$ $=38$
A_2	12×8 $+14$ $=110$	6×6 $+8$ $=44$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times$ $[3+1]$ $=36$	$5\times$ $[3+3]$ $=30$
A_2	$12\times$ $[2+1]$ $=36$	$6\times$ $[2+3]$ $=30$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$86-36$ $=50$	$38-30$ $=8$
A_2	$110-36$ $=74$	$44-30$ $=14$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	50	8
A_2	74	14



4.6 Data-Service Contract (C1–C3)

A data-services company must choose standard or automated processing before a contract is confirmed as standard-volume or high-volume. Work estimates assign standard processing 4 batches under standard volume and 7 under high volume, while automation handles 5 and 9 batches, respectively.

The method agreement pays fixed revenue of 55 thousand USD for standard processing or 72 for automation. The client adds a service-level payment of 16 thousand USD in the standard-volume state and 9 in the high-volume state. Processing costs 6 thousand USD per batch with the standard method or 4 with automation, and implementation adds a method-based fixed cost of 18 thousand USD or 31, respectively. The company must keep both revenue payments fixed while applying only the processing rate to batch quantity.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Use standard data processing	A_2: Use automated data processing
States of nature	S_1: Standard-volume contract	S_2: High-volume contract
Uncertain event	Contract requirements confirmed after the processing-method decision	
Consequences	Profit in thousand USD for each processing-method–contract-state pair	

Revenue contains alternative-based and state-based fixed components:

$$R(A_i, S_j) = f_r(A_i) + f_r(S_j).$$

Cost contains alternative-based variable and fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c(A_i) + f_c(A_i).$$

Therefore,

$$P(A_i, S_j) = [f_r(A_i) + f_r(S_j)] - [q(A_i, S_j)c(A_i) + f_c(A_i)]\,.$$

C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use standard data processing.
A_2	—	Use automated data processing.
S_1	—	Standard-volume contract.
S_2	—	High-volume contract.
$f_r(A_1)$	55 thousand USD	Method agreement payment for A_1 (fixed amount).
$f_r(S_1)$	16 thousand USD	Service-level payment for S_1 (fixed amount).
$q(A_1, S_1)$	4 processing batches	Forecast quantity for A_1 under S_1.
$c(A_1)$	6 thousand USD per processing batch	Processing rate for A_1 (rate per unit).
$f_c(A_1)$	18 thousand USD	Implementation cost for A_1 (fixed amount).
$f_r(S_2)$	9 thousand USD	Service-level payment for S_2 (fixed amount).
$q(A_1, S_2)$	7 processing batches	Forecast quantity for A_1 under S_2.
$f_r(A_2)$	72 thousand USD	Method agreement payment for A_2 (fixed amount).
$q(A_2, S_1)$	5 processing batches	Forecast quantity for A_2 under S_1.
$c(A_2)$	4 thousand USD per processing batch	Processing rate for A_2 (rate per unit).
$f_c(A_2)$	31 thousand USD	Implementation cost for A_2 (fixed amount).
$q(A_2, S_2)$	9 processing batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$55+16$ $=71$	$55+9$ $=64$
A_2	$72+16$ $=88$	$72+9$ $=81$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(A_1)$ $+f_c(A_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times c(A_1)$ $+f_c(A_1)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(A_2)$ $+f_c(A_2)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times c(A_2)$ $+f_c(A_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	4×6 $+18$ $=42$	7×6 $+18$ $=60$
A_2	5×4 $+31$ $=51$	9×4 $+31$ $=67$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$71-42$ $=29$	$64-60$ $=4$
A_2	$88-51$ $=37$	$81-67$ $=14$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	29	4
A_2	37	14



5 Three-Component Models

Problems:

5.1. Regional Parcel Delivery (C1–C3)

5.2. Solar Installation Programme (C1–C3)

5.3. Hybrid Conference Production (C1–C3)

5.4. Cold-Storage Export Contract (C1–C3)

5.1 Regional Parcel Delivery (C1–C3)

A regional logistics company must select its standard or express delivery plan before conditions become high demand with light traffic or moderate demand with heavy traffic. Forecasts give the standard plan 8 hundreds of delivery batches in the first state and 5 in the second, and the express plan 10 and 6 hundreds, respectively.

The standard service earns 9 thousand USD per hundred batches and express service earns 11; demand conditions add 3 per hundred batches under high demand or 2 under moderate demand. Service agreements also pay a fixed 16 thousand USD for the standard plan and 24 for express. Operating costs are 4 thousand USD per hundred batches for standard and 5 for express, while traffic adds 1 per hundred batches when light and 3 when heavy. Distribution arrangements finally cost a fixed 10 thousand USD in the high-demand, light-traffic state and 15 in the moderate-demand, heavy-traffic state.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Use the standard delivery plan	A_2**:** Use the express delivery plan
States of nature	S_1**:** High demand with light traffic	S_2**:** Moderate demand with heavy traffic
Uncertain event	Demand and traffic conditions revealed after the delivery-plan decision	
Consequences	Profit in thousand USD for each delivery-plan–state pair	

Revenue contains alternative-based and state-based variable components and an alternative-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i).$$

Cost contains alternative-based and state-based variable components and a state-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \{q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i)\} \\ & - \{q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(S_j)\}\,. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use the standard delivery plan.
A_2	—	Use the express delivery plan.
S_1	—	High demand with light traffic.
S_2	—	Moderate demand with heavy traffic.
$q(A_1, S_1)$	8 hundreds of delivery batches	Forecast quantity for A_1 under S_1.
$r(A_1)$	9 thousand USD per hundred delivery batches	Delivery service rate for A_1 (rate per unit).
$r(S_1)$	3 thousand USD per hundred delivery batches	Demand supplement for S_1 (rate per unit).
$f_r(A_1)$	16 thousand USD	Service agreement payment for A_1 (fixed amount).
$c(A_1)$	4 thousand USD per hundred delivery batches	Operating rate for A_1 (rate per unit).
$c(S_1)$	1 thousand USD per hundred delivery batches	Traffic-related rate for S_1 (rate per unit).
$f_c(S_1)$	10 thousand USD	Distribution arrangement cost for S_1 (fixed amount).
$q(A_1, S_2)$	5 hundreds of delivery batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	2 thousand USD per hundred delivery batches	Demand supplement for S_2 (rate per unit).
$c(S_2)$	3 thousand USD per hundred delivery batches	Traffic-related rate for S_2 (rate per unit).
$f_c(S_2)$	15 thousand USD	Distribution arrangement cost for S_2 (fixed amount).
$q(A_2, S_1)$	10 hundreds of delivery batches	Forecast quantity for A_2 under S_1.
$r(A_2)$	11 thousand USD per hundred delivery batches	Delivery service rate for A_2 (rate per unit).
$f_r(A_2)$	24 thousand USD	Service agreement payment for A_2 (fixed amount).
$c(A_2)$	5 thousand USD per hundred delivery batches	Operating rate for A_2 (rate per unit).
$q(A_2, S_2)$	6 hundreds of delivery batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[9+3]$ $+16$ $=112$	$5\times$ $[9+2]$ $+16$ $=71$
A_2	$10\times$ $[11+3]$ $+24$ $=164$	$6\times$ $[11+2]$ $+24$ $=102$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[4+1]$ $+10$ $=50$	$5\times$ $[4+3]$ $+15$ $=50$
A_2	$10\times$ $[5+1]$ $+10$ $=70$	$6\times$ $[5+3]$ $+15$ $=63$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$112-50$ $=62$	$71-50$ $=21$
A_2	$164-70$ $=94$	$102-63$ $=39$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	62	21
A_2	94	39

5.2 Solar Installation Programme (C1–C3)

A renewable-energy company must offer a standard-panel or high-efficiency-panel programme before demand and permits resolve into strong commercial demand or limited residential demand. It forecasts 7 installation batches for standard panels in the strong state and 4 in the limited state, compared with 6 and 3 batches for high-efficiency panels.

Standard panels earn 12 thousand USD per batch and high-efficiency panels 18, while the market adds 4 per batch in the strong state or 2 in the limited state. An incentive provides fixed revenue of 20 thousand USD under strong commercial demand and 11 under limited residential demand. Installation inputs cost 6 thousand USD per standard-panel batch or 8 per high-efficiency batch. Programme setup costs a fixed 18 thousand USD for standard panels or 25 for high-efficiency panels, and permits add fixed costs of 9 in the strong state or 14 in the limited state.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Offer the standard-panel programme	A_2: Offer the high-efficiency-panel programme
States of nature	S_1: Strong commercial demand	S_2: Limited residential demand
Uncertain event	Customer demand and permit conditions revealed after the programme decision	
Consequences	Profit in thousand USD for each programme–market-state pair	

Revenue contains alternative-based and state-based variable components and a state-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(S_j).$$

Cost contains an alternative-based variable component and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c(A_i) + f_c(A_i) + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \{q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(S_j)\} \\ & - \{q(A_i, S_j)c(A_i) + f_c(A_i) + f_c(S_j)\}\,. \end{aligned}$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Offer the standard-panel programme.
A_2	—	Offer the high-efficiency-panel programme.
S_1	—	Strong commercial demand.
S_2	—	Limited residential demand.
$q(A_1, S_1)$	7 installation batches	Forecast quantity for A_1 under S_1.
$r(A_1)$	12 thousand USD per installation batch	Panel installation revenue rate for A_1 (rate per unit).
$r(S_1)$	4 thousand USD per installation batch	Market supplement for S_1 (rate per unit).
$f_r(S_1)$	20 thousand USD	Incentive payment for S_1 (fixed amount).
$c(A_1)$	6 thousand USD per installation batch	Installation input rate for A_1 (rate per unit).
$f_c(A_1)$	18 thousand USD	Programme setup cost for A_1 (fixed amount).
$f_c(S_1)$	9 thousand USD	Permit cost for S_1 (fixed amount).
$q(A_1, S_2)$	4 installation batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	2 thousand USD per installation batch	Market supplement for S_2 (rate per unit).
$f_r(S_2)$	11 thousand USD	Incentive payment for S_2 (fixed amount).
$f_c(S_2)$	14 thousand USD	Permit cost for S_2 (fixed amount).
$q(A_2, S_1)$	6 installation batches	Forecast quantity for A_2 under S_1.
$r(A_2)$	18 thousand USD per installation batch	Panel installation revenue rate for A_2 (rate per unit).
$c(A_2)$	8 thousand USD per installation batch	Installation input rate for A_2 (rate per unit).
$f_c(A_2)$	25 thousand USD	Programme setup cost for A_2 (fixed amount).
$q(A_2, S_2)$	3 installation batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$7\times$ $[12+4]$ $+20$ $=132$	$4\times$ $[12+2]$ $+11$ $=67$
A_2	$6\times$ $[18+4]$ $+20$ $=152$	$3\times$ $[18+2]$ $+11$ $=71$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(A_1)$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times c(A_1)$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(A_2)$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times c(A_2)$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	7×6 $+18+9$ $=69$	4×6 $+18+14$ $=56$
A_2	6×8 $+25+9$ $=82$	3×8 $+25+14$ $=63$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$132-69$ $=63$	$67-56$ $=11$
A_2	$152-82$ $=70$	$71-63$ $=8$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	63	11
A_2	70	8



5.3 Hybrid Conference Production (C1–C3)

An events company must choose a standard or immersive hybrid-conference package before attendance and sponsorship are known to be strong or limited. Production forecasts allocate 6 event segments to the standard package in the strong state and 4 in the limited state, while the immersive package uses 8 and 5 segments, respectively.

Each standard segment earns 13 thousand USD and each immersive segment 16. Booking contracts add fixed revenue of 14 thousand USD for the standard package or 22 for the immersive package, while sponsors add a further fixed 18 in the strong state or 9 in the limited state. Delivery costs 5 thousand USD per segment under strong conditions and 7 under limited conditions. Package setup adds a fixed cost of 24 for standard or 37 for immersive, and licensing adds a state-based fixed cost of 8 or 13, respectively.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Use the standard hybrid package	A_2**:** Use the immersive hybrid package
States of nature	S_1**:** Strong attendance and sponsorship	S_2**:** Limited attendance and sponsorship
Uncertain event	Attendance and sponsorship conditions revealed after the package decision	
Consequences	Profit in thousand USD for each production-package–state pair	

Revenue contains an alternative-based variable component and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r(A_i) + f_r(A_i) + f_r(S_j).$$

Cost contains a state-based variable component and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c(S_j) + f_c(A_i) + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = \; & \{q(A_i, S_j)r(A_i) + f_r(A_i) + f_r(S_j)\} \\ & - \{q(A_i, S_j)c(S_j) + f_c(A_i) + f_c(S_j)\}. \end{aligned}$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use the standard hybrid package.
A_2	—	Use the immersive hybrid package.
S_1	—	Strong attendance and sponsorship.
S_2	—	Limited attendance and sponsorship.
$q(A_1, S_1)$	6 event segments	Forecast quantity for A_1 under S_1.
$r(A_1)$	13 thousand USD per event segment	Segment revenue rate for A_1 (rate per unit).
$f_r(A_1)$	14 thousand USD	Booking contract payment for A_1 (fixed amount).
$f_r(S_1)$	18 thousand USD	Sponsorship payment for S_1 (fixed amount).
$c(S_1)$	5 thousand USD per event segment	Segment delivery rate for S_1 (rate per unit).
$f_c(A_1)$	24 thousand USD	Package setup cost for A_1 (fixed amount).
$f_c(S_1)$	8 thousand USD	Licensing cost for S_1 (fixed amount).
$q(A_1, S_2)$	4 event segments	Forecast quantity for A_1 under S_2.
$f_r(S_2)$	9 thousand USD	Sponsorship payment for S_2 (fixed amount).
$c(S_2)$	7 thousand USD per event segment	Segment delivery rate for S_2 (rate per unit).
$f_c(S_2)$	13 thousand USD	Licensing cost for S_2 (fixed amount).
$q(A_2, S_1)$	8 event segments	Forecast quantity for A_2 under S_1.
$r(A_2)$	16 thousand USD per event segment	Segment revenue rate for A_2 (rate per unit).
$f_r(A_2)$	22 thousand USD	Booking contract payment for A_2 (fixed amount).
$f_c(A_2)$	37 thousand USD	Package setup cost for A_2 (fixed amount).
$q(A_2, S_2)$	5 event segments	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(A_1)$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times r(A_1)$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(A_2)$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times r(A_2)$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×13 $+14+18$ $=110$	4×13 $+14+9$ $=75$
A_2	8×16 $+22+18$ $=168$	5×16 $+22+9$ $=111$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times c(S_1)$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times c(S_2)$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times c(S_1)$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times c(S_2)$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	6×5 $+24+8$ $=62$	4×7 $+24+13$ $=65$
A_2	8×5 $+37+8$ $=85$	5×7 $+37+13$ $=85$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$110-62$ $=48$	$75-65$ $=10$
A_2	$168-85$ $=83$	$111-85$ $=26$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	48	10
A_2	83	26



5.4 Cold-Storage Export Contract (C1–C3)

A food-export company must choose a modular or large cold-storage facility before export demand is strong or limited. Capacity and demand forecasts give the modular facility 9 storage batches under strong demand and 5 under limited demand, and the large facility 12 and 7 batches, respectively.

Export conditions set revenue at 10 thousand USD per batch under strong demand or 7 under limited demand. The chosen capacity earns fixed revenue of 20 thousand USD for the modular facility or 30 for the large facility, and the export programme adds fixed revenue of 16 in the strong state or 8 in the limited state. Operating rates are 4 thousand USD per modular batch and 3 per large-facility batch; energy adds 2 per batch under strong demand or 4 under limited demand. Preparing the configuration also costs a fixed 28 thousand USD for modular storage or 42 for large storage.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Use the modular cold-storage facility	A_2: Use the large cold-storage facility
States of nature	S_1: Strong export demand	S_2: Limited export demand
Uncertain event	Export demand and energy-market conditions revealed after the facility decision	
Consequences	Profit in thousand USD for each storage-facility–state pair	

Revenue contains a state-based variable component and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r(S_j) + f_r(A_i) + f_r(S_j).$$

Cost contains alternative-based and state-based variable components and an alternative-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \{q(A_i, S_j)r(S_j) + f_r(A_i) + f_r(S_j)\} \\ & - \{q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i)\}\,. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use the modular cold-storage facility.
A_2	—	Use the large cold-storage facility.
S_1	—	Strong export demand.
S_2	—	Limited export demand.
$q(A_1, S_1)$	9 storage batches	Forecast quantity for A_1 under S_1.
$r(S_1)$	10 thousand USD per storage batch	Export revenue rate for S_1 (rate per unit).
$f_r(A_1)$	20 thousand USD	Capacity agreement payment for A_1 (fixed amount).
$f_r(S_1)$	16 thousand USD	Export programme payment for S_1 (fixed amount).
$c(A_1)$	4 thousand USD per storage batch	Facility operating rate for A_1 (rate per unit).
$c(S_1)$	2 thousand USD per storage batch	Energy rate for S_1 (rate per unit).
$f_c(A_1)$	28 thousand USD	Configuration preparation cost for A_1 (fixed amount).
$q(A_1, S_2)$	5 storage batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	7 thousand USD per storage batch	Export revenue rate for S_2 (rate per unit).
$f_r(S_2)$	8 thousand USD	Export programme payment for S_2 (fixed amount).
$c(S_2)$	4 thousand USD per storage batch	Energy rate for S_2 (rate per unit).
$q(A_2, S_1)$	12 storage batches	Forecast quantity for A_2 under S_1.
$f_r(A_2)$	30 thousand USD	Capacity agreement payment for A_2 (fixed amount).
$c(A_2)$	3 thousand USD per storage batch	Facility operating rate for A_2 (rate per unit).
$f_c(A_2)$	42 thousand USD	Configuration preparation cost for A_2 (fixed amount).
$q(A_2, S_2)$	7 storage batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times r(S_1)$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times r(S_2)$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times r(S_1)$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times r(S_2)$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	9×10 $+20+16$ $=126$	5×7 $+20+8$ $=63$
A_2	12×10 $+30+16$ $=166$	7×7 $+30+8$ $=87$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(A_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(A_1)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(A_2)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(A_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$9\times$ $[4+2]$ $+28$ $=82$	$5\times$ $[4+4]$ $+28$ $=68$
A_2	$12\times$ $[3+2]$ $+42$ $=102$	$7\times$ $[3+4]$ $+42$ $=91$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$126-82$ $=44$	$63-68$ $=-5$
A_2	$166-102$ $=64$	$87-91$ $=-4$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	44	-5
A_2	64	-4

The negative payoffs under limited export demand represent losses because the revenue earned does not cover the variable and fixed costs.



6 Complete Four-Component Models

Problems:

6.1. Smart Manufacturing Contract (C1–C3)

6.2. Distribution Centre Automation (C1–C3)

6.3. Electric Vehicle Charging Network (C1–C3)

6.1 Smart Manufacturing Contract (C1–C3)

A manufacturer must select conventional or automated assembly before conditions become strong demand with favourable inputs or limited demand with expensive inputs. Forecast production is 8 batches for conventional assembly in the favourable state and 5 in the expensive-input state, versus 11 and 6 batches for automation.

Conventional batches earn a system-based 12 thousand USD each and automated batches 15; market conditions add 4 per batch in the favourable state or 2 in the expensive-input state. The contract also pays fixed revenue of 18 thousand USD for conventional assembly or 28 for automation, while a market incentive adds fixed revenue of 14 or 7 in the respective states.

Operating rates are 5 thousand USD per conventional batch and 6 per automated batch. Materials add 2 per batch under favourable inputs or 4 under expensive inputs. Setup costs are fixed at 24 thousand USD for conventional assembly and 38 for automation, while compliance adds a fixed 9 in the favourable state or 15 in the expensive-input state.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Use the conventional assembly system	A_2**:** Use the automated assembly system
States of nature	S_1**:** Strong demand with favourable input conditions	S_2**:** Limited demand with expensive inputs
Uncertain event	Demand and input-market conditions revealed after the production-system decision	
Consequences	Profit in thousand USD for each production-system–state pair	

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = & \{q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j)\} \\ & - \{q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j)\}\,. \end{aligned}$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use the conventional assembly system.
A_2	—	Use the automated assembly system.
S_1	—	Strong demand with favourable input conditions.
S_2	—	Limited demand with expensive inputs.
$q(A_1, S_1)$	8 production batches	Forecast quantity for A_1 under S_1.
$r(A_1)$	12 thousand USD per production batch	Assembly revenue rate for A_1 (rate per unit).
$r(S_1)$	4 thousand USD per production batch	Market supplement for S_1 (rate per unit).
$f_r(A_1)$	18 thousand USD	Contract payment for A_1 (fixed amount).
$f_r(S_1)$	14 thousand USD	Market incentive for S_1 (fixed amount).
$c(A_1)$	5 thousand USD per production batch	Assembly operating rate for A_1 (rate per unit).
$c(S_1)$	2 thousand USD per production batch	Material rate for S_1 (rate per unit).
$f_c(A_1)$	24 thousand USD	System setup cost for A_1 (fixed amount).
$f_c(S_1)$	9 thousand USD	Compliance cost for S_1 (fixed amount).
$q(A_1, S_2)$	5 production batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	2 thousand USD per production batch	Market supplement for S_2 (rate per unit).
$f_r(S_2)$	7 thousand USD	Market incentive for S_2 (fixed amount).
$c(S_2)$	4 thousand USD per production batch	Material rate for S_2 (rate per unit).
$f_c(S_2)$	15 thousand USD	Compliance cost for S_2 (fixed amount).
$q(A_2, S_1)$	11 production batches	Forecast quantity for A_2 under S_1.
$r(A_2)$	15 thousand USD per production batch	Assembly revenue rate for A_2 (rate per unit).
$f_r(A_2)$	28 thousand USD	Contract payment for A_2 (fixed amount).
$c(A_2)$	6 thousand USD per production batch	Assembly operating rate for A_2 (rate per unit).
$f_c(A_2)$	38 thousand USD	System setup cost for A_2 (fixed amount).
$q(A_2, S_2)$	6 production batches	Forecast quantity for A_2 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[12+4]$ $+18+14$ $=160$	$5\times$ $[12+2]$ $+18+7$ $=95$
A_2	$11\times$ $[15+4]$ $+28+14$ $=251$	$6\times$ $[15+2]$ $+28+7$ $=137$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[5+2]$ $+24+9$ $=89$	$5\times$ $[5+4]$ $+24+15$ $=84$
A_2	$11\times$ $[6+2]$ $+38+9$ $=135$	$6\times$ $[6+4]$ $+38+15$ $=113$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$160-89$ $=71$	$95-84$ $=11$
A_2	$251-135$ $=116$	$137-113$ $=24$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	71	11
A_2	116	24

6.2 Distribution Centre Automation (C1–C3)

A retailer must choose manual, semi-automated, or robotic distribution before order volume and energy prices resolve into high volume with favourable energy or moderate volume with expensive energy. Forecast quantities, in hundreds of order batches, are 8 and 5 for manual distribution, 11 and 7 for semi-automation, and 14 and 8 for robotics across those respective states.

The configurations earn 12, 15, and 18 thousand USD per hundred order batches, respectively, because their service capabilities differ. The market adds 4 per hundred batches under favourable energy or 2 under expensive energy. Service contracts provide fixed revenue of 16, 24, and 35 thousand USD for manual, semi-automated, and robotic distribution, respectively; delivery incentives add fixed revenue of 15 in the high-volume state or 8 in the moderate-volume state.

Operating rates are 5, 6, and 8 thousand USD per hundred batches for the three respective configurations. Energy adds 2 per hundred batches in the favourable state or 4 in the expensive state. Installation creates fixed costs of 22, 34, and 50 thousand USD for manual, semi-automated, and robotic distribution, respectively, while compliance adds fixed costs of 9 under high volume or 16 under moderate volume.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Use the manual distribution configuration	A_2: Use the semi-automated distribution configuration	A_3: Use the robotic distribution configuration
States of nature	S_1: High order volume with favourable energy prices	S_2: Moderate order volume with expensive energy	
Uncertain event	Order volume and energy-market conditions revealed after the distribution-configuration decision		
Consequences	Profit in thousand USD for each distribution-configuration–state pair		

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \{q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j)\} \\ & - \{q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j)\}\,. \end{aligned}$$



C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Use the manual distribution configuration.
A_2	—	Use the semi-automated distribution configuration.
A_3	—	Use the robotic distribution configuration.
S_1	—	High order volume with favourable energy prices.
S_2	—	Moderate order volume with expensive energy.
$q(A_1, S_1)$	8 hundreds of order batches	Forecast quantity for A_1 under S_1.
$r(A_1)$	12 thousand USD per hundred order batches	Distribution service rate for A_1 (rate per unit).
$r(S_1)$	4 thousand USD per hundred order batches	Market supplement for S_1 (rate per unit).
$f_r(A_1)$	16 thousand USD	Service contract payment for A_1 (fixed amount).
$f_r(S_1)$	15 thousand USD	Delivery incentive for S_1 (fixed amount).
$c(A_1)$	5 thousand USD per hundred order batches	Configuration operating rate for A_1 (rate per unit).
$c(S_1)$	2 thousand USD per hundred order batches	Energy rate for S_1 (rate per unit).
$f_c(A_1)$	22 thousand USD	Installation cost for A_1 (fixed amount).
$f_c(S_1)$	9 thousand USD	Compliance cost for S_1 (fixed amount).
$q(A_1, S_2)$	5 hundreds of order batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	2 thousand USD per hundred order batches	Market supplement for S_2 (rate per unit).
$f_r(S_2)$	8 thousand USD	Delivery incentive for S_2 (fixed amount).
$c(S_2)$	4 thousand USD per hundred order batches	Energy rate for S_2 (rate per unit).
$f_c(S_2)$	16 thousand USD	Compliance cost for S_2 (fixed amount).
$q(A_2, S_1)$	11 hundreds of order batches	Forecast quantity for A_2 under S_1.
$r(A_2)$	15 thousand USD per hundred order batches	Distribution service rate for A_2 (rate per unit).
$f_r(A_2)$	24 thousand USD	Service contract payment for A_2 (fixed amount).
$c(A_2)$	6 thousand USD per hundred order batches	Configuration operating rate for A_2 (rate per unit).
$f_c(A_2)$	34 thousand USD	Installation cost for A_2 (fixed amount).
$q(A_2, S_2)$	7 hundreds of order batches	Forecast quantity for A_2 under S_2.
$q(A_3, S_1)$	14 hundreds of order batches	Forecast quantity for A_3 under S_1.
$r(A_3)$	18 thousand USD per hundred order batches	Distribution service rate for A_3 (rate per unit).
$f_r(A_3)$	35 thousand USD	Service contract payment for A_3 (fixed amount).
$c(A_3)$	8 thousand USD per hundred order batches	Configuration operating rate for A_3 (rate per unit).
$f_c(A_3)$	50 thousand USD	Installation cost for A_3 (fixed amount).
$q(A_3, S_2)$	8 hundreds of order batches	Forecast quantity for A_3 under S_2.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$
A_3	$q(A_3,S_1)\times$ $[r(A_3)+r(S_1)]$ $+f_r(A_3)+f_r(S_1)$ $=R(A_3,S_1)$	$q(A_3,S_2)\times$ $[r(A_3)+r(S_2)]$ $+f_r(A_3)+f_r(S_2)$ $=R(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[12+4]$ $+16+15$ $=159$	$5\times$ $[12+2]$ $+16+8$ $=94$
A_2	$11\times$ $[15+4]$ $+24+15$ $=248$	$7\times$ $[15+2]$ $+24+8$ $=151$
A_3	$14\times$ $[18+4]$ $+35+15$ $=358$	$8\times$ $[18+2]$ $+35+8$ $=203$

Cost

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$
A_3	$q(A_3,S_1)\times$ $[c(A_3)+c(S_1)]$ $+f_c(A_3)+f_c(S_1)$ $=C(A_3,S_1)$	$q(A_3,S_2)\times$ $[c(A_3)+c(S_2)]$ $+f_c(A_3)+f_c(S_2)$ $=C(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$8\times$ $[5+2]$ $+22+9$ $=87$	$5\times$ $[5+4]$ $+22+16$ $=83$
A_2	$11\times$ $[6+2]$ $+34+9$ $=131$	$7\times$ $[6+4]$ $+34+16$ $=120$
A_3	$14\times$ $[8+2]$ $+50+9$ $=199$	$8\times$ $[8+4]$ $+50+16$ $=162$

Profit

Symbolic calculations.

Alternative	S_1	S_2
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$
A_3	$R(A_3,S_1)-C(A_3,S_1)$ $=P(A_3,S_1)$	$R(A_3,S_2)-C(A_3,S_2)$ $=P(A_3,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$159-87$ $=72$	$94-83$ $=11$
A_2	$248-131$ $=117$	$151-120$ $=31$
A_3	$358-199$ $=159$	$203-162$ $=41$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2
A_1	72	11
A_2	117	31
A_3	159	41



6.3 Electric Vehicle Charging Network (C1–C3)

An energy company must build a neighbourhood or rapid-charging network before utilisation and electricity prices become high and favourable, moderate and standard, or low and expensive. Forecast quantities, in hundreds of charging-session batches, are 10, 7, and 4 for the neighbourhood network across those states, and 14, 9, and 5 for the rapid network.

Neighbourhood service earns a network-based 11 thousand USD per hundred charging-session batches and rapid charging earns 16. Utilisation conditions add 4, 3, and 2 thousand USD per hundred batches in the high, moderate, and low states, respectively. Network service agreements provide fixed revenue of 18 thousand USD for the neighbourhood network or 30 for rapid charging, while utilisation incentives add fixed revenue of 16, 10, and 5 in the respective states.

Equipment costs 4 thousand USD per hundred batches for the neighbourhood network or 7 for rapid charging. Electricity adds 2, 3, and 5 thousand USD per hundred batches in the respective states. Infrastructure creates a fixed cost of 25 thousand USD for the neighbourhood design or 45 for rapid charging, while state-dependent maintenance adds fixed costs of 8, 12, and 18, respectively. The company must preserve all eight components when computing each network–state profit.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Build the neighbourhood charging network	A_2: Build the rapid-charging network	
States of nature	S_1: High utilisation with favourable electricity prices	S_2: Moderate utilisation with standard electricity prices	S_3: Low utilisation with expensive electricity
Uncertain event	Charging demand and electricity-market conditions revealed after the network-design decision		
Consequences	Profit in thousand USD for each charging-network–state pair		

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \{q(A_i, S_j)\,[r(A_i) + r(S_j)] + f_r(A_i) + f_r(S_j)\} \\ & - \{q(A_i, S_j)\,[c(A_i) + c(S_j)] + f_c(A_i) + f_c(S_j)\}\,. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

1. Symbols, values, and meanings.

Symbol	Value	Meaning in this problem.
A_1	—	Build the neighbourhood charging network.
A_2	—	Build the rapid-charging network.
S_1	—	High utilisation with favourable electricity prices.
S_2	—	Moderate utilisation with standard electricity prices.
S_3	—	Low utilisation with expensive electricity.
$q(A_1, S_1)$	10 hundreds of charging-session batches	Forecast quantity for A_1 under S_1.
$r(A_1)$	11 thousand USD per hundred charging-session batches	Charging service rate for A_1 (rate per unit).
$r(S_1)$	4 thousand USD per hundred charging-session batches	Utilisation supplement for S_1 (rate per unit).
$f_r(A_1)$	18 thousand USD	Network agreement payment for A_1 (fixed amount).
$f_r(S_1)$	16 thousand USD	Utilisation incentive for S_1 (fixed amount).
$c(A_1)$	4 thousand USD per hundred charging-session batches	Equipment rate for A_1 (rate per unit).
$c(S_1)$	2 thousand USD per hundred charging-session batches	Electricity rate for S_1 (rate per unit).
$f_c(A_1)$	25 thousand USD	Infrastructure cost for A_1 (fixed amount).
$f_c(S_1)$	8 thousand USD	Maintenance cost for S_1 (fixed amount).
$q(A_1, S_2)$	7 hundreds of charging-session batches	Forecast quantity for A_1 under S_2.
$r(S_2)$	3 thousand USD per hundred charging-session batches	Utilisation supplement for S_2 (rate per unit).
$f_r(S_2)$	10 thousand USD	Utilisation incentive for S_2 (fixed amount).
$c(S_2)$	3 thousand USD per hundred charging-session batches	Electricity rate for S_2 (rate per unit).
$f_c(S_2)$	12 thousand USD	Maintenance cost for S_2 (fixed amount).
$q(A_1, S_3)$	4 hundreds of charging-session batches	Forecast quantity for A_1 under S_3.
$r(S_3)$	2 thousand USD per hundred charging-session batches	Utilisation supplement for S_3 (rate per unit).
$f_r(S_3)$	5 thousand USD	Utilisation incentive for S_3 (fixed amount).
$c(S_3)$	5 thousand USD per hundred charging-session batches	Electricity rate for S_3 (rate per unit).
$f_c(S_3)$	18 thousand USD	Maintenance cost for S_3 (fixed amount).
$q(A_2, S_1)$	14 hundreds of charging-session batches	Forecast quantity for A_2 under S_1.
$r(A_2)$	16 thousand USD per hundred charging-session batches	Charging service rate for A_2 (rate per unit).
$f_r(A_2)$	30 thousand USD	Network agreement payment for A_2 (fixed amount).
$c(A_2)$	7 thousand USD per hundred charging-session batches	Equipment rate for A_2 (rate per unit).
$f_c(A_2)$	45 thousand USD	Infrastructure cost for A_2 (fixed amount).
$q(A_2, S_2)$	9 hundreds of charging-session batches	Forecast quantity for A_2 under S_2.
$q(A_2, S_3)$	5 hundreds of charging-session batches	Forecast quantity for A_2 under S_3.

2. Symbolic and numerical calculations.

All calculated amounts below are in thousand USD. The row and column indices refer to the input table.

Revenue

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times$ $[r(A_1)+r(S_1)]$ $+f_r(A_1)+f_r(S_1)$ $=R(A_1,S_1)$	$q(A_1,S_2)\times$ $[r(A_1)+r(S_2)]$ $+f_r(A_1)+f_r(S_2)$ $=R(A_1,S_2)$	$q(A_1,S_3)\times$ $[r(A_1)+r(S_3)]$ $+f_r(A_1)+f_r(S_3)$ $=R(A_1,S_3)$
A_2	$q(A_2,S_1)\times$ $[r(A_2)+r(S_1)]$ $+f_r(A_2)+f_r(S_1)$ $=R(A_2,S_1)$	$q(A_2,S_2)\times$ $[r(A_2)+r(S_2)]$ $+f_r(A_2)+f_r(S_2)$ $=R(A_2,S_2)$	$q(A_2,S_3)\times$ $[r(A_2)+r(S_3)]$ $+f_r(A_2)+f_r(S_3)$ $=R(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$10\times$ $[11+4]$ $+18+16$ $=184$	$7\times$ $[11+3]$ $+18+10$ $=126$	$4\times$ $[11+2]$ $+18+5$ $=75$
A_2	$14\times$ $[16+4]$ $+30+16$ $=326$	$9\times$ $[16+3]$ $+30+10$ $=211$	$5\times$ $[16+2]$ $+30+5$ $=125$

Cost

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$q(A_1,S_1)\times$ $[c(A_1)+c(S_1)]$ $+f_c(A_1)+f_c(S_1)$ $=C(A_1,S_1)$	$q(A_1,S_2)\times$ $[c(A_1)+c(S_2)]$ $+f_c(A_1)+f_c(S_2)$ $=C(A_1,S_2)$	$q(A_1,S_3)\times$ $[c(A_1)+c(S_3)]$ $+f_c(A_1)+f_c(S_3)$ $=C(A_1,S_3)$
A_2	$q(A_2,S_1)\times$ $[c(A_2)+c(S_1)]$ $+f_c(A_2)+f_c(S_1)$ $=C(A_2,S_1)$	$q(A_2,S_2)\times$ $[c(A_2)+c(S_2)]$ $+f_c(A_2)+f_c(S_2)$ $=C(A_2,S_2)$	$q(A_2,S_3)\times$ $[c(A_2)+c(S_3)]$ $+f_c(A_2)+f_c(S_3)$ $=C(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$10\times$ $[4+2]$ $+25+8$ $=93$	$7\times$ $[4+3]$ $+25+12$ $=86$	$4\times$ $[4+5]$ $+25+18$ $=79$
A_2	$14\times$ $[7+2]$ $+45+8$ $=179$	$9\times$ $[7+3]$ $+45+12$ $=147$	$5\times$ $[7+5]$ $+45+18$ $=123$

Profit

Symbolic calculations.

Alternative	S_1	S_2	S_3
A_1	$R(A_1,S_1)-C(A_1,S_1)$ $=P(A_1,S_1)$	$R(A_1,S_2)-C(A_1,S_2)$ $=P(A_1,S_2)$	$R(A_1,S_3)-C(A_1,S_3)$ $=P(A_1,S_3)$
A_2	$R(A_2,S_1)-C(A_2,S_1)$ $=P(A_2,S_1)$	$R(A_2,S_2)-C(A_2,S_2)$ $=P(A_2,S_2)$	$R(A_2,S_3)-C(A_2,S_3)$ $=P(A_2,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
A_1	$184-93$ $=91$	$126-86$ $=40$	$75-79$ $=-4$
A_2	$326-179$ $=147$	$211-147$ $=64$	$125-123$ $=2$

3. Final payoff table: profit (thousand USD).

Alternative	S_1	S_2	S_3
A_1	91	40	-4
A_2	147	64	2

The negative payoff for the neighbourhood charging network under low utilisation represents a loss because the revenue earned does not cover the variable and fixed costs.